DGMS 2019 GEOLOGY • QUESTION 42

Dana Chemical Mineralogy: Halides, Hydroxides, Carbonates & Phosphates

Topic: Mineral Classification & Systematic Mineralogy | Primary Reference: Dexter Perkins (Mineralogy, 3rd ed., Appendix & Ch. 2)

Question Statement

Q.42. Match List-I (Mineral name) with List-II (Mineral classification):

List-I (Mineral name):
A. Sylvite
B. Manganite
C. Siderite
D. Monazite

List-II (Mineral classification):
1. Carbonate
2. Phosphate
3. Hydroxide
4. Halide

(a) A-4, B-3, C-1, D-2
(b) A-1, B-3, C-4, D-2
(c) A-4, B-2, C-3, D-1
(d) A-3, B-2, C-4, D-1

a **A-4, B-3, C-1, D-2 (Correct Answer)**

b A-1, B-3, C-4, D-2

c A-4, B-2, C-3, D-1

d A-3, B-2, C-4, D-1

✓ **Correct Answer: (a) A-4, B-3, C-1, D-2 [Sylvite = Halide; Manganite = Hydroxide; Siderite = Carbonate; Monazite = Phosphate].**

Textbook Evidence: Dana's Chemical Classification System

Dana Mineral Systematics: Matching Minerals to Chemical Classes (Q.42)

Mineral Name	Chemical Formula	Diagnostic Anion Group	Dana Mineral Class	Key Geological Occurrenc
Sylvite	KCl	Cl^- (Halogen anion)	**4. Halide**	Marine evaporite deposits
Manganite	MnO(OH)	OH^- & O^{2-} (Hydroxyl / o	**3. Hydroxide**	Hydrothermal veins & sed
Siderite	$FeCO_3$	$[CO_3]^{2-}$ (Carbonate grou	**1. Carbonate**	Banded iron formations, c
Monazite	$(Ce,La,Nd,Th)PO_4$	$[PO_4]^{3-}$ (Phosphate grou	**2. Phosphate**	Heavy mineral beach plac

Figure 42.1: Dana Classification of Non-Silicate Minerals (Dexter Perkins, *Mineralogy*, 3rd ed., Chapter 2 & Appendix). Classification based on the dominant anionic group: Halides (Cl^-, F^-), Hydroxides ($(\text{OH})^-$), Carbonates ($[\text{CO}_3]^{2-}$), and Phosphates ($[\text{PO}_4]^{3-}$).

> *"In the Dana System of Mineralogy, minerals are grouped primarily according to the dominant anion or anionic complex in their chemical formula:*
> • **Halides:** *Contain halogen anions (Cl^-, F^-, Br^-, I^-) bonded to metals (e.g., halite NaCl, sylvite KCl, fluorite CaF_2).*
> • **Hydroxides / Oxy-hydroxides:** *Contain hydroxyl $(\text{OH})^-$ groups (e.g., brucite Mg(OH)_2, manganite MnO(OH), goethite FeO(OH)).*
> • **Carbonates:** *Contain planar $[\text{CO}_3]^{2-}$ complex anions (e.g., calcite CaCO_3, siderite FeCO_3, dolomite).*
> • **Phosphates:** *Contain tetrahedral $[\text{PO}_4]^{3-}$ complex anions (e.g., apatite $\text{Ca}_5(\text{PO}_4)_3(\text{F,Cl,OH})$, monazite $(\text{Ce,La,Th})\text{PO}_4$)."*
>
> **— Dexter Perkins, *Mineralogy*, Chapter 2 (Classification of Minerals), pp. 47–55**

Step-by-Step Chemical Matching

A. Sylvite → 4. Halide **[MATCH 1]**

Formula & Chemistry: KCl (Potassium chloride). Belongs to the isometric halite structure group. The dominant anion is chloride (Cl^-), making sylvite a primary member of the **Halide** class. It is the premier commercial ore of potash, precipitating in late-stage evaporite marine basins.

B. Manganite → 3. Hydroxide **[MATCH 2]**

Formula & Chemistry: MnO(OH) (Manganese oxy-hydroxide). Contains essential structural hydroxyl $(\text{OH})^-$ groups coordinated with Mn^{3+} cations. In Dana's taxonomy, oxy-hydroxides are classified under **Hydroxides** (Oxides and Hydroxides class). It occurs in low-temperature hydrothermal veins and secondary manganese replacement deposits.

C. Siderite → 1. Carbonate **[MATCH 3]**

Formula & Chemistry: FeCO_3 (Iron(II) carbonate). Belongs to the rhombohedral calcite group ($R\bar{3}c$). The fundamental anionic complex is $[\text{CO}_3]^{2-}$. Siderite is an important iron ore found in sedimentary iron formations (clay ironstone), coal beds, and hydrothermal veins.

D. Monazite → 2. Phosphate **[MATCH 4]**

Formula & Chemistry: $(\text{Ce},\text{La},\text{Nd},\text{Th})\text{PO}_4$ (Rare-earth phosphate). Contains tetrahedral $[\text{PO}_4]^{3-}$ groups coordinating large light rare-earth elements (LREEs) and thorium. Monazite is the primary source of cerium, thorium, and rare earths, occurring in granites, pegmatites, and concentrated as heavy mineral beach sands (placers).

Overview: Dana Chemical Mineral Classes

Mineral Class	Defining Anion / Radical	Key Examples	Diagnostic Properties
Native Elements	None (pure elements)	Gold (Au), Diamond/ Graphite (C), Sulfur ($ \text{S}$)	Metallic or covalent bonding
Sulfides	S^{2-} (Sulfide ion)	Galena (PbS), Sphalerite (ZnS), Pyrite (FeS_2)	High density, metallic luster, dark streak
Halides	**$\text{F}^-, \text{Cl}^-, \text{Br}^-, \text{I}^-$**	**Sylvite (KCl)**, Halite (NaCl), Fluorite ($ \text{CaF}_2$)	Ionic bonding, low hardness, salty/bitter taste, water solubility
Hydroxides	**$(\text{OH})^-$ (Hydroxyl ion)**	**Manganite (MnO(OH))**, Brucite, Goethite, Gibbsite	Dehydrate upon heating to form oxides
Carbonates	**$[\text{CO}_3]^{2-}$**	**Siderite (FeCO_3)**, Calcite (CaCO_3), Dolomite	Effervesce in dilute $ \text{HCl}$, high birefringence, rhombohedral cleavage
Phosphates	**$[\text{PO}_4]^{3-}$**	**Monazite ($(\text{Ce,La,Th})\text{PO}_4$)**, Apatite ($ \text{Ca}_5(\text{PO}_4)_3\text{F}$)	High refractive index, moderate-to-high density, radioactive when Th/U-rich

Key Takeaway & Exam Strategy

Quick Matching Logic:

- **Sylvite (KCl):** $\text{Cl}^- \rightarrow$ **Halide (4)**.
- **Manganite (MnO(OH)):** $\text{OH}^- \rightarrow$ **Hydroxide (3)**.
- **Siderite (FeCO_3):** $\text{CO}_3^{2-} \rightarrow$ **Carbonate (1)**.
- **Monazite ($(\text{Ce,La,Th})\text{PO}_4$):** $\text{PO}_4^{3-} \rightarrow$ **Phosphate (2)**.

- Direct matching: **A-4, B-3, C-1, D-2** (Option a).